riskdep: a crisis on a mainnet fork

Belem RWA EURCV vault · 2-minute demo · scripts/fork/demo.sh --short

Why a mainnet fork

A local copy of Ethereum at block 26,165,572 (anvil)

Real vault, real positions, real oracles and roles

The crisis is caused by roles the protocols grant

- Issuer's feed admin, Midas timelock, a liquidator
- No exploit; nothing reaches the real network

riskdep watches the vault live, block by block

The vault: three fragilities

- Belem RWA EURCV: 4.71M EURCV, curated by RockawayX
- 86 % lent in one market, against mGLOeuro (LLTV 77 %)
- One EOA writes the mGLOeuro price (NAV)
  - Fixed band 90,000 - 110,000 EUR; below needs new code
  - New code passes a 48 h timelock
- One borrower holds all 3.68M EURCV of debt, LTV 0.70

What each mGLOeuro price means

mGLOeuro price (EUR)	Change	What happens
100,000	today	Borrower LTV 0.70: safe
90,857	-9.1 %	Borrower becomes liquidatable (LTV 77 %)
90,000	-10 %	Lowest price the feed accepts today
75,142	-24.9 %	Below this, liquidation leaves bad debt
70,000	-30 %	251,761 EURCV of bad debt for the vault

Prediction, before the crisis

Markdown -10 %

Vault loss: 0 EURCV
Borrower liquidatable
Collateral still covers the debt
Verdict: STAY

Markdown -30 %

Vault loss: 251,761 EURCV (5.34 %)
Largest depositor: 250,399
Leaving first saves only 382
Verdict: STAY

What riskdep finds at start-up

- Feed band: mGLOeuro cannot fall below 90,000 without an upgrade
  - A larger loss stays invisible to the market until then
- Timelock: the upgrade must pass Midas's 48 h timelock
  - riskdep watches its queue
- Exit gates abdicated: the curator can never lock withdrawals

The crisis, act by act

- Issuer sets the NAV to 90,000: one transaction, no delay
- Real loss is 30 %: Midas schedules a feed upgrade
- 48 hours later (fork clock moved): upgrade executed, NAV 70,000
- A liquidator seizes the rest: bad debt written off

What riskdep says, and when

Moment	Alert	Level
NAV -10 %	Oracle moved -10 %; liquidation curve	warn
Upgrade scheduled	Executable in 48 h: the window to act	warn
Upgrade executed	Feed code changed	critical
NAV 70,000	Unrealised bad debt 252,236 EURCV	critical
Liquidation	Bad debt realised 252,914 EURCV	critical

Predicted against observed

Predicted before the crisis: 251,761 EURCV
Observed on the fork: 252,914 EURCV

Gap 0.5 %: interest accrued during the 48 h jump

What the demo proves

Brief's criteria

- Discovery: feed, its admin, band, timelock
- Impact: 251,761 predicted, confirmed
- Decision: 48 h warning; STAY explained
- Real data: mainnet state at a real block

Limits

- Crisis is played, not observed
- Feed upgrade is demo code
- Keyless upstreams can throttle
- One vault, one scenario

Run it

scripts/fork/demo.sh --short (about 2 minutes)
scripts/fork/demo.sh --record (asciinema video)
scripts/fork/demo.sh --stop (stop everything)

Offline, no fork:
riskdep watch --vault 0xD6d4… --replay --record-as fork-belem